

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week -3

Scenario: BookFlow wants to capture user reviews and technical specs for books, where every book has different details. A fixed SQL table is too restrictive, so MongoDB is used to manage this unstructured content and generate a "Top Rated" report using the aggregation pipeline.

Tasks: 1) Document Modeling — create a book_metadata collection with embedded patterns (reviews list inside the book document) and referenced patterns (linking to member_id in SQL). 2) Advanced Querying — use comparison operators (\$gt, \$in) to find books rated above 4, and a regex query to find books with "Digital" formats. 3) Aggregation Pipeline — match books published after 2020, \$unwind the reviews array, then \$group by title and compute the average rating. 4) Indexing for Speed — create a text index on the review comments field and use .explain() to check whether the query uses the index.

Task 1: Document Modelling

Sample document

```
Atlas atlas-3orha6-shard-0 [primary] schoolDB> use bookflow;
switched to db bookflow
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.insertOne({ title: "The Silent Algorithm", author: "R. Menon",
| genre: "Technology", publishedYear: 2022,
| formats: ["Hardcover", "Digital", "Audiobook"], specs: {
| pages: 340,
| fileSize_MB: 4.2, drm: true, narrator: "K. Rao", runtime_min: 610
| },
| avgRating: 4.6, reviews: [
| {
| member_id: "M1001", rating: 5,
| comment: "Excellent deep dive into digital systems design.", date: ISODate("2023-01-15")
| },
| {
| member_id: "M1042", rating: 4,
| comment: "Good but the Digital edition had formatting issues.", date: ISODate("2023-02-20")
| }
| ]
| });
|
| {
| acknowledged: true,
| insertedId: ObjectId('6a893180553d2fc70d47a21c')
| }
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.insertMany([
| {
| title: "Ocean of Code", author: "A. Fernandes", genre: "Technology", publishedYear: 2021,
| formats: ["Digital", "Paperback"],
| specs: { pages: 210, fileSize_MB: 2.1, drm: false }, avgRating: 4.2,
| reviews: [
| { member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-10") },
| { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date: ISODate("2022-06-01") }
| ],
| },
| {
| title: "Whispers of Yesterday", author: "S. Kapoor",
```

```

| title: "Ocean of Code", author: "A. Fernandes", genre: "Technology", publishedYear: 2021,
| formats: ["Digital", "Paperback"],
| specs: { pages: 210, fileSize_MB: 2.1, drm: false }, avgRating: 4.2,
| reviews: [
| { member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-10") },
| { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date: ISODate("2022-06-01") }
| ]
| },
| {
| title: "Whispers of Yesterday", author: "S. Kapoor",
| genre: "Fiction", publishedYear: 2019,
|
| formats: ["Hardcover", "Paperback"], specs: { pages: 410, bindbinding: "Sewn" }, avgRating: 3.8,
| reviews: [
| { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
| { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
| ]
| },
| {
| title: "Quantum Horizons", author: "R. Menon", genre: "Science", publishedYear: 2023,
| formats: ["Digital", "Audiobook"],
| specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 }, avgRating: 4.9,
| reviews: [
| { member_id: "M4001", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
| { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
| ]
| }
| ]);
|
| {
| acknowledged: true,
| insertedIds: {
|   '0': ObjectId('6a8931a9553d2fc70d47a21d'),
|   '1': ObjectId('6a8931a9553d2fc70d47a21e'),
|   '2': ObjectId('6a8931a9553d2fc70d47a21f')
| }
| }
| }

```

Task 2: Advanced Querying

\$gt (greater than)

```
db.book_metadata.find({ avgRating: { $gt: 4 } });
```

.

\$in (set membership)

```
db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } });
```

```
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } });
[
  {
    _id: ObjectId('6a893180553d2fc70d47a21c'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8931a9553d2fc70d47a21d'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
  }
]
```

```

publishedYear: 2021,
formats: [ 'Digital', 'Paperback' ],
specs: { pages: 210, fileSize_MB: 2.1, drm: false },
avgRating: 4.2,
reviews: [
  {
    member_id: 'M2001',
    rating: 4,
    comment: 'Solid Digital read, clean layout.',
    date: ISODate('2022-05-10T00:00:00.000Z')
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: ISODate('2022-06-01T00:00:00.000Z')
  }
]
},
{
  _id: ObjectId('6a8931a9553d2fc70d47a21f'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}

```

Regex matching

```
db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } });
```

```

    date: ISODate('2023-09-14T00:00:00.000Z')
  }
]
}
]
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } });
[
  {
    _id: ObjectId('6a893180553d2fc70d47a21c'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8931a9553d2fc70d47a21d'),
    title: 'Ocean of Code',

```

```

author: 'A. Fernandes',
genre: 'Technology',
publishedYear: 2021,
formats: [ 'Digital', 'Paperback' ],
specs: { pages: 210, fileSize_MB: 2.1, drm: false },
avgRating: 4.2,
reviews: [
  {
    member_id: 'M2001',
    rating: 4,
    comment: 'Solid Digital read, clean layout.',
    date: ISODate('2022-05-10T00:00:00.000Z')
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: ISODate('2022-06-01T00:00:00.000Z')
  }
]
},
{
  _id: ObjectId('6a8931a9553d2fc70d47a21f'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,

```

```

    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}

```

Why this matters for BookFlow

These operators let the application answer real product questions directly at the database layer — "show me highly rated books," "show me anything available digitally," "find reviews mentioning digital issues" — without pulling every document into application code and filtering there, which would not scale.

a) Books rated above 4 (\$gt)

```

]
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { avgRating: { $gt: 4 } },
| { title: 1, avgRating: 1, _id: 0 }
| );
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]

```

b) Books available in specific formats (\$in)

```

]
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { formats: { $in: ["Digital", "Audiobook"] } },
| { title: 1, formats: 1, _id: 0 }
| );
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]

```

c) Combine both with \$and (implicit)

```

]
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find({ avgRating: { $gt: 4 }, formats: { $in: ["Digital"] }
| });
[
  {
    _id: ObjectId('6a893180553d2fc70d47a21c'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8931a9553d2fc70d47a21d'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',

```

```

genre: 'Technology',
publishedYear: 2021,
formats: [ 'Digital', 'Paperback' ],
specs: { pages: 210, fileSize_MB: 2.1, drm: false },
avgRating: 4.2,
reviews: [
  {
    member_id: 'M2001',
    rating: 4,
    comment: 'Solid Digital read, clean layout.',
    date: ISODate('2022-05-10T00:00:00.000Z')
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: ISODate('2022-06-01T00:00:00.000Z')
  }
],
},
{
  _id: ObjectId('6a8931a9553d2fc70d47a21f'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]

```

d) Regex query — books with "Digital" format

```
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { formats: { $regex: "Digital", $options: "i" } },
| { title: 1, formats: 1 }
| );
|
| [
|   {
|     _id: ObjectId('6a893180553d2fc70d47a21c'),
|     title: 'The Silent Algorithm',
|     formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
|   },
|   {
|     _id: ObjectId('6a8931a9553d2fc70d47a21d'),
|     title: 'Ocean of Code',
|     formats: [ 'Digital', 'Paperback' ]
|   },
|   {
|     _id: ObjectId('6a8931a9553d2fc70d47a21f'),
|     title: 'Quantum Horizons',
|     formats: [ 'Digital', 'Audiobook' ]
|   }
| ]
```

e) Regex query — reviews mentioning "Digital" in comments (nested field)

```
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } },
| { title: 1, "reviews.$": 1 }
| );
|
| [
|   {
|     _id: ObjectId('6a893180553d2fc70d47a21c'),
|     title: 'The Silent Algorithm',
|     reviews: [
|       {
|         member_id: 'M1001',
|         rating: 5,
|         comment: 'Excellent deep dive into digital systems design.',
|         date: ISODate('2023-01-15T00:00:00.000Z')
|       }
|     ]
|   },
|   {
|     _id: ObjectId('6a8931a9553d2fc70d47a21d'),
|     title: 'Ocean of Code',
|     reviews: [
|       {
|         member_id: 'M2001',
|         rating: 4,
|         comment: 'Solid Digital read, clean layout.',
|         date: ISODate('2022-05-10T00:00:00.000Z')
|       }
|     ]
|   },
|   {
|     _id: ObjectId('6a8931a9553d2fc70d47a21f'),
|     title: 'Quantum Horizons',
|     reviews: [
|       {
|         member_id: 'M4001',
|         rating: 5,
|         comment: 'The Digital format made annotations so easy.',
|         date: ISODate('2023-08-01T00:00:00.000Z')
|       }
|     ]
|   }
| ]
```

Task 3: Aggregation Pipeline — "Top Rated" Report

Stage-by-stage breakdown

1. \$match — filter early

```
{ $match: { publishedYear: { $gt: 2020 } } }
```

2. \$unwind — flatten the array

```
{ $unwind: "$reviews" }
```

3. \$group — aggregate per title

```
{
  $group: {
    _id: "$title",
    avgReviewRating: { $avg: "$reviews.rating" },
    totalReviews: { $sum: 1 },
    author: { $first: "$author" }
  }
}
```

4. \$sort — order by rating

```
{ $sort: { avgReviewRating: -1 } }
```

5. \$project — reshape output

```
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: { $round: ["$avgReviewRating", 2] },
    totalReviews: 1
  }
}
```

```

Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.aggregate([
| // Stage 1: filter books published after 2020
| { $match: { publishedYear: { $gt: 2020 } } },
|
| // Stage 2: deconstruct the reviews array - one output doc per review
| { $unwind: "$reviews" },
|
| // Stage 3: group by title, compute average rating & review count
| {
|   $group: {
|     _id: "$title",
|     avgReviewRating: { $avg: "$reviews.rating" }, totalReviews: { $sum: 1 },
|     author: { $first: "$author" }
|   }
| },
|
| // Stage 4: sort descending by average rating
| { $sort: { avgReviewRating: -1 } },
|
| // Stage 5: reshape output
| {
|   $project: {
|     _id: 0,
|     title: "$_id", author: 1,
|     avgReviewRating: { $round: ["$avgReviewRating", 2] }, totalReviews: 1
|   }
| }
| ]);
|
| {
|   totalReviews: 2,
|   author: 'R. Menon',

```

```
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  }
]
```

Task 4: Indexing for Speed

Text index

```
db.book_metadata.createIndex({ "reviews.comment": "text" });
```

```
db.book_metadata.find({ $text: { $search: "Digital" } });
```

.

Verifying index usage with .explain()

```
db.book_metadata.find({ $text: { $search: "Digital" } }).explain("executionStats");
```

Why regex can't use a standard index efficiently

```
db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" }  
}).explain("executionStats");
```

Multikey indexes

```
db.book_metadata.createIndex({ formats: 1 });
```

a) Create a text index on review comments

```
]
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.createIndex({ "reviews.comment": "text" });
reviews.comment_text
Atlas atlas-3orha6-shard-0 [primary] bookflow>
```

b) Query using the text index

```
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { $text: { $search: "Digital" } },
| { title: 1, score: { $meta: "textScore" } }
| ).sort({ score: { $meta: "textScore" } });
|
| [
|   {
|     _id: ObjectId('6a893180553d2fc70d47a21c'),
|     title: 'The Silent Algorithm',
|     score: 1.1833333333333333
|   },
|   {
|     _id: ObjectId('6a8931a9553d2fc70d47a21f'),
|     title: 'Quantum Horizons',
|     score: 0.6
|   },
|   {
|     _id: ObjectId('6a8931a9553d2fc70d47a21d'),
|     title: 'Ocean of Code',
|     score: 0.6
|   }
| ]
```

c) Verify index usage with .explain()

```

Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { $text: { $search: "Digital" } }
| ).explain("executionStats");
|
| {
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'bookflow.book_metadata',
|     parsedQuery: {
|       '$text': {
|         '$search': 'Digital',
|         '$language': 'english',
|         '$caseSensitive': false,
|         '$diacriticSensitive': false
|       }
|     },
|     indexFilterSet: false,
|     queryHash: 'CF6F4CEE',
|     planCacheShapeHash: 'CF6F4CEE',
|     planCacheKey: '08852285',
|     optimizationTimeMillis: 0,
|     maxIndexedOrSolutionsReached: false,
|     maxIndexedAndSolutionsReached: false,
|     maxScansToExplodeReached: false,
|     prunedSimilarIndexes: false,
|     winningPlan: {
|       isCached: false,
|       stage: 'TEXT_MATCH',
|       indexPrefix: {},
|       indexName: 'reviews.comment_text',
|       parsedTextQuery: {
|         terms: [ 'digit' ],
|         negatedTerms: [],
|         phrases: [],
|         negatedPhrases: []
|       }
|     },
|     textIndexVersion: 3,

```

```
inputStage: {
  stage: 'FETCH',
  inputStage: {
    stage: 'IXSCAN',
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 0,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    indexPrefix: {},
  }
}
```

```
,
textIndexVersion: 3,
docsRejected: 0,
inputStage: {
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 3,
```

```

        indexBounds: {},
        keysExamined: 3,
        seeks: 1,
        dupsTested: 3,
        dupsDropped: 0
    }
}
},
queryShapeHash: '46EF773C046D1CE446CA691E7FBCE3DCB3ED54D15BA162EC9BD029701553687E',
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'Digital' } },
  '$db': 'bookflow'
},
serverInfo: {
  host: 'ac-mtx6id4-shard-00-01.rpjfl0n.mongoddb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787559130, i: 8 }),
  signature: {

```

```

    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1787559130, i: 8 }),
    signature: {
      hash: Binary.createFromBase64('3Gv+tHUmPIELzWTpk3LgNLAZzoU=', 0),
      keyId: Long('7641639996687908885')
    }
  },
  operationTime: Timestamp({ t: 1787559130, i: 8 })
}

```

d) Compare against a non-indexed regex scan

```
db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } }
).explain("executionStats");
```

```
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } }
| ).explain("executionStats");
|
| {
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'bookflow.book_metadata',
|     parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
|     indexFilterSet: false,
|     queryHash: '9DF7CB16',
|     planCacheShapeHash: '9DF7CB16',
|     planCacheKey: 'A9183C86',
|     optimizationTimeMillis: 0,
|     maxIndexedOrSolutionsReached: false,
|     maxIndexedAndSolutionsReached: false,
|     maxScansToExplodeReached: false,
|     prunedSimilarIndexes: false,
|     winningPlan: {
|       isCached: false,
|       stage: 'COLLSCAN',
|       filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
|       direction: 'forward'
|     },
|     rejectedPlans: []
|   },
|   executionStats: {
|     executionSuccess: true,
|     nReturned: 3,
|     executionTimeMillis: 0,
|     totalKeysExamined: 0,
|     totalDocsExamined: 4,
|     executionStages: {
|       isCached: false,
|       stage: 'COLLSCAN',
|       filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
|       nReturned: 3,
```

```

    direction: 'forward',
    docsExamined: 4
  },
},
queryShapeHash: 'B327CE1453AC3D30B57392893CEE0F4DF8BE56201D345EE00E2AB7084FEE6EE8',
command: {
  find: 'book_metadata',
  filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
  '$db': 'bookflow'
},
serverInfo: {
  host: 'ac-mtx6id4-shard-00-01.rpjl0n.mongoddb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787559400, i: 3 }),
  signature: {
    hash: Binary.createFromBase64('rE0+8wVkJLUj9nujDEPGhMXLq/c=', 0),
    keyId: Long('7641639996687908885')
  }
},
operationTime: Timestamp({ t: 1787559400, i: 3 })

```

This will show stage: "COLLSCAN" — a regex without an anchored prefix cannot use a standard index, illustrating why the text index is preferable for keyword search on comments.

e) Supporting index for common queries (multikey index)

Since formats and reviews are arrays, an index on them becomes a **multikey index** automatically:

```
db.book_metadata.createIndex({ formats: 1 });

db.book_metadata.createIndex({ avgRating: -1 });

db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats");

// winningPlan.stage should now show IXSCAN instead of COLLSCAN
```

```
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.createIndex({ avgRating: -1 });
avgRating_-1
Atlas atlas-3orha6-shard-0 [primary] bookflow> db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: '85BA8276',
    planCacheShapeHash: '85BA8276',
    planCacheKey: '6EB4A3FD',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf.0, 4)' ] }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
```

```
executionSuccess: true,
nReturned: 3,
executionTimeMillis: 1,
totalKeysExamined: 3,
totalDocsExamined: 3,
executionStages: {
  isCached: false,
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 1,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 1,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    keyPattern: { avgRating: -1 },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: { avgRating: [] },
    isUnique: false,
    isSparse: false,
    isPartial: false,
```

```

    }
  }
},
queryShapeHash: '6AB2391EEC16606F1308C88C345B1A4B2637B4FCE78BE101E8CFC829A0D6B506',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$db': 'bookflow'
},
serverInfo: {
  host: 'ac-mtx6id4-shard-00-01.rpjl0n.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
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  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
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  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787559554, i: 13 }),
  signature: {
    hash: Binary.createFromBase64('o/cwt707NumirzaB4b+Xw5pQRtk=', 0),
    keyId: Long('7641639996687908885')
  }
},
operationTime: Timestamp({ t: 1787559554, i: 13 })
}

```